

By using cloud services, the risk of data loss and hardware failure is reduced. In contrast to the traditional on-premise approach, cloud hosting companies employ advanced techniques of encryption and authentication.

- b) **Scalability and savings:** Cloud computing platforms offer an enhanced level of scalability compared to the traditional on-premise approach. They offer dynamic addition and release of storage space as well as computing power during peak loads. In the traditional approach, if there is a shortage of computing resources, the user has to purchase the hardware infrastructure.
- c) **Governed access:** In cloud computing, the practice of data governance ensures regulated and monitored access to confidential data. This feature also supports access to data anytime and from anywhere. This is achieved by implementing data governance protocols (authorisations, policies, metadata, etc) in a cloud database system. Thus, by using on-demand access to data and resources, cloud computing offers businesses significant cost savings.
- d) **Automated backups and disaster recovery:** In the traditional approach, a disaster recovery system requires constant maintenance and support. Disaster recovery in cloud computing entails storing critical data and applications in cloud storage and failover to a secondary site in case of a disaster.

Evolving a management strategy for data on the cloud

In the IT industry, cloud data management strategies are critical to achieving corporate goals. The steps below outline how to handle data in a cloud data server.

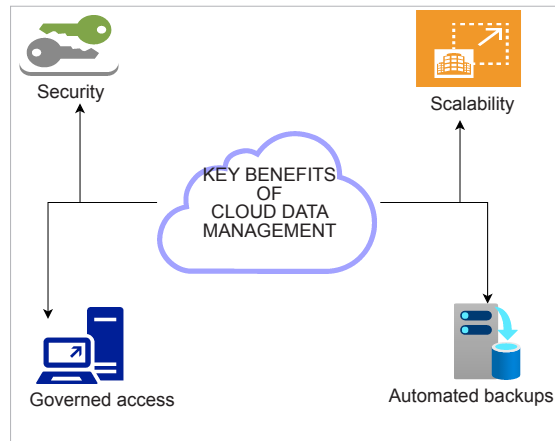


Figure 1: Key benefits of cloud data management

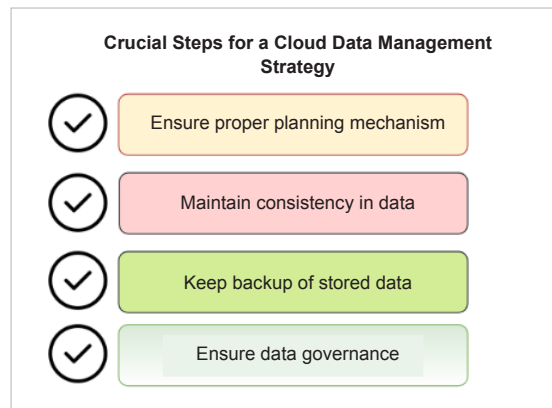


Figure 2: Crucial steps for cloud data management

- a) **Ensure a proper planning mechanism:** Planning is the key to project management success. In cloud data management, for successful dumping of data into the cloud environment the user has to take care of issues such as: i)

ensuring that complete data is offloaded to the cloud, ii) implementing proper mechanisms for efficient data access, iii) deciding how data processing will be done at the cloud server.

b) **Maintain consistency in data:** Data consistency ensures that information stored in the database is accurate and that there is no redundancy in the data.

c) **Keep a backup of stored data:** Cloud service providers follow the practice of regular backup. In a cloud data centre, regular backup is done by copying data to a remote location server.

d) **Ensure data governance:** Governance protocols are implemented to ensure regulated and monitored access to data.

With the growth in the size of databases, today's data managers need a system that can provide ease of access and efficient management of data from any location. To achieve this objective, cloud data management strategies have been widely adopted across multiple industries. **END** 🐧

Reference

- [1] <https://www.talend.com/resources/what-is-cloud-data-management/>
- [2] <https://www.xplenty.com/blog/cloud-data-management-guide/>
- [3] <https://blog.panoply.io/data-management-strategy-and-best-practices-in-the-cloud>
- [4] <https://aws.amazon.com/blogs/storage/tag/data-management/>

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